The Hong Kong University of Science and Technology
IEDA 1250, Summer 2026
Mid-term Exam Solution

Student ID ________________ Name: ________________________________

Please read the following instructions carefully.

- You have **90 minutes** to complete this exam. This question booklet contains 6 questions (including a bonus question) for the total of 120 points. Check to see if any pages are missing.
- Read the instructions for individual questions carefully before answering the questions.

Question	Points	Score
1	10	
2	20	
3	20	
4	20	
5	20	
6	30	
Total	120	

1. **(10 points)** Solve the following linear program (LP) graphically. Clearly identify the feasible region, the optimal solution, and the optimal objective value.

$$\begin{aligned} \max_{x_1,x_2} \quad & x_1 + \frac{1}{2}x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \le 6, \\ & x_1 + 2x_2 \ge 0, \\ & 2x_1 - x_2 \le 4, \\ & 2x_1 - x_2 \ge -2. \end{aligned}$$

Solution: The feasible region is a diamond with vertices

$$\left(-\frac{4}{5}, \frac{2}{5}\right), \quad \left(\frac{2}{5}, \frac{14}{5}\right), \quad \left(\frac{14}{5}, \frac{8}{5}\right), \quad \left(\frac{8}{5}, -\frac{4}{5}\right).$$

Now evaluate the objective function

$$z = x_1 + \frac{1}{2}x_2$$

at the four vertices:

Vertex	$z = x_1 + \frac{1}{2}x_2$
$\left(-\frac{4}{5}, \frac{2}{5}\right)$	$-\frac{3}{5}$
$\left(\frac{8}{5}, -\frac{4}{5}\right)$	$\frac{7}{5}$
$\left(\frac{2}{5}, \frac{14}{5}\right)$	$\frac{9}{5}$
$\left(\frac{14}{5}, \frac{8}{5}\right)$	$\frac{18}{5}$

Therefore, the optimal solution is

$$\boxed{x_1^* = \frac{14}{5}, \qquad x_2^* = \frac{8}{5}}$$

and the optimal objective value is

$$\boxed{z^* = \frac{18}{5}.}$$

2. **(20 points)** Consider the following primal LP:

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 + x_3 + x_4 + x_5 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 + x_4 \le 3, \\ & x_1 - x_2 + 2x_3 + x_5 = 7, \\ & x_1 + 2x_2 - x_3 + 3x_4 + x_5 \le 6, \\ & x_2 + x_3 - x_4 + 2x_5 = 7, \\ & x_1 \ge 0, \quad x_2 \ge 0, \end{aligned}$$

where x_3, x_4, x_5 are unrestricted in sign. Suppose that y_1, y_2, y_3, and y_4 are the dual variables corresponding to the four constraints of the primal LP, respectively, and that the optimal dual solution is

$$(y_1^*, y_2^*, y_3^*, y_4^*) = (2, -1, 0, 1).$$

Use the complementary slackness conditions to find an optimal primal solution.

Solution: Given the primal LP and the optimal dual solution

$$(y_1^*, y_2^*, y_3^*, y_4^*) = (2, -1, 0, 1),$$

we use complementary slackness to find an optimal primal solution.

The dual LP is

$$\begin{array}{ll} \min & 3y_1 + 7y_2 + 6y_3 + 7y_4 \\ \text{s.t.} & 2y_1 + y_2 + y_3 \geq 2, \\ & y_1 - y_2 + 2y_3 + y_4 \geq 3, \\ & y_1 + 2y_2 - y_3 + y_4 = 1, \\ & y_1 + 3y_3 - y_4 = 1, \\ & y_2 + y_3 + 2y_4 = 1, \\ & y_1 \geq 0, \quad y_3 \geq 0, \quad y_2, y_4 \text{ free.} \end{array}$$

Step 1: Apply complementary slackness.

For primal constraints:

$$y_1^* \left(3 - (2x_1 + x_2 + x_3 + x_4)\right) = 0.$$

Since $y_1^* = 2 \neq 0$, the first primal constraint must be tight:

$$2x_1 + x_2 + x_3 + x_4 = 3.$$

Also,

$$y_3^* \left(6 - (x_1 + 2x_2 - x_3 + 3x_4 + x_5)\right) = 0.$$

Since $y_3^* = 0$, this gives no additional equality condition.

For dual constraints corresponding to x_1 and x_2:

$$x_1 \left(2y_1^* + y_2^* + y_3^* - 2\right) = 0.$$

Substituting the dual solution gives

$$2(2) + (-1) + 0 - 2 = 1 \neq 0.$$

Therefore,

$$x_1 = 0.$$

Similarly,

$$x_2 \left(y_1^* - y_2^* + 2y_3^* + y_4^* - 3\right) = 0.$$

Substituting the dual solution gives

$$2 - (-1) + 2(0) + 1 - 3 = 1 \neq 0.$$

Therefore,

$$x_2 = 0.$$

Step 2: Substitute $x_1 = 0$ and $x_2 = 0$ into the primal constraints.

The tight first primal constraint gives

$$x_3 + x_4 = 3.$$

The second primal constraint gives

$$2x_3 + x_5 = 7.$$

The fourth primal constraint gives

$$x_3 - x_4 + 2x_5 = 7.$$

Therefore, we solve

$$\begin{cases} x_3 + x_4 = 3, \\ 2x_3 + x_5 = 7, \\ x_3 - x_4 + 2x_5 = 7. \end{cases}$$

Solving this system, we obtain

$$x_3 = 2, \qquad x_4 = 1, \qquad x_5 = 3.$$

Hence, an optimal primal solution is

$$\boxed{x_1^* = 0, \quad x_2^* = 0, \quad x_3^* = 2, \quad x_4^* = 1, \quad x_5^* = 3.}$$

The optimal primal objective value is

$$2x_1^* + 3x_2^* + x_3^* + x_4^* + x_5^* = 0 + 0 + 2 + 1 + 3 = 6.$$

Thus,

$$\boxed{z^* = 6.}$$

3. **(20 points)** A relief agency needs to ship medicine boxes from two warehouses to three clinics. Warehouse 1 has 20 boxes available, and Warehouse 2 has 30 boxes available. Clinic 1 needs 10 boxes, Clinic 2 needs 25 boxes, and Clinic 3 needs 15 boxes. Since total supply equals total demand, all boxes must be shipped. The shipping cost per box is shown below.

	Clinic 1	Clinic 2	Clinic 3
Warehouse 1	4	6	8
Warehouse 2	5	4	3

Let x_{ij} denote the number of boxes shipped from Warehouse i to Clinic j.

(a) Formulate a linear program that minimizes the total shipping cost while satisfying all clinic demands.

(b) Write down the dual linear program. Let u_1, u_2 be the dual variables associated with the two warehouse constraints, and let v_1, v_2, v_3 be the dual variables associated with the three clinic constraints.

(c) Suppose the optimal shipment plan is

$$x_{11}^* = 10, \quad x_{12}^* = 10, \quad x_{13}^* = 0, \quad x_{21}^* = 0, \quad x_{22}^* = 15, \quad x_{23}^* = 15.$$

Also suppose an optimal dual solution is

$$u_1^* = 2, \quad u_2^* = 0, \quad v_1^* = 2, \quad v_2^* = 4, \quad v_3^* = 3.$$

Explain the meaning of the dual variables intuitively. What is the economic interpretation of the dual LP and its optimal solution?

Solution: Let x_{ij} denote the number of boxes shipped from Warehouse i to Clinic j.

(a) Primal LP formulation.

The objective is to minimize the total shipping cost:

$$\min \quad 4x_{11} + 6x_{12} + 8x_{13} + 5x_{21} + 4x_{22} + 3x_{23}.$$

Subject to the warehouse supply constraints:

$$x_{11} + x_{12} + x_{13} = 20,$$

$$x_{21} + x_{22} + x_{23} = 30,$$

and the clinic demand constraints:

$$x_{11} + x_{21} = 10,$$

$$x_{12} + x_{22} = 25,$$

$$x_{13} + x_{23} = 15.$$

Also,

$$x_{ij} \geq 0, \qquad i = 1, 2, \quad j = 1, 2, 3.$$

(b) Dual LP formulation.

Let u_1, u_2 be the dual variables associated with the two warehouse constraints, and let v_1, v_2, v_3 be the dual variables associated with the three clinic constraints. Since all primal constraints are equality constraints, all dual variables are unrestricted in sign.

The dual LP is

$$\max \quad 20u_1 + 30u_2 + 10v_1 + 25v_2 + 15v_3$$

subject to

$$u_1 + v_1 \leq 4,$$

$$u_1 + v_2 \leq 6,$$

$$u_1 + v_3 \leq 8,$$

$$u_2 + v_1 \leq 5,$$

$$u_2 + v_2 \leq 4,$$

$$u_2 + v_3 \leq 3,$$

where

$$u_1, u_2, v_1, v_2, v_3 \text{ are free.}$$

(c) Economic interpretation of the dual variables.

The dual variables can be interpreted as shadow prices. The variables u_1 and u_2 represent the marginal values associated with the supplies available at Warehouses 1 and 2, respectively. The variables v_1, v_2, v_3 represent the marginal values associated with satisfying demands at Clinics 1, 2, and 3, respectively.

The dual constraint

$$u_i + v_j \leq c_{ij}$$

means that the combined value of one box supplied from Warehouse i and delivered to Clinic j cannot exceed the actual shipping cost c_{ij}. Therefore, the dual LP tries to assign values to warehouse supplies and clinic demands so that these values are consistent with all shipping costs.

Using the given optimal dual solution

$$u_1^* = 2, \qquad u_2^* = 0, \qquad v_1^* = 2, \qquad v_2^* = 4, \qquad v_3^* = 3,$$

the dual objective value is

$$20(2) + 30(0) + 10(2) + 25(4) + 15(3) = 40 + 0 + 20 + 100 + 45 = 205.$$

The cost of the given primal shipment plan is

$$4(10) + 6(10) + 8(0) + 5(0) + 4(15) + 3(15) = 40 + 60 + 0 + 0 + 60 + 45 = 205.$$

Since the primal and dual objective values are equal, the given shipment plan and the given dual solution are both optimal.

Economically, the optimal dual solution provides a system of prices that supports the optimal transportation plan. Routes with positive shipments satisfy

$$u_i^* + v_j^* = c_{ij},$$

while routes with zero shipments satisfy

$$u_i^* + v_j^* \leq c_{ij}.$$

Thus, only routes whose shipping cost is justified by the corresponding warehouse and clinic shadow prices are used in the optimal plan.